

2018 Ph H2 Q7

Section: Particles and Waves

Topic: Photoelectric effect

(a) UV light shines on a negatively charged plate but no discharge occurs. Suggest a reason.

(b)(i) Define the work function of $6.94 \times 10^{-19} \text{ J}$.

(ii) If irradiance is reduced, what happens to maximum KE of photoelectrons? Justify.

(c) Add line to KE vs frequency graph for aluminium with threshold $9.0 \times 10^{14} \text{ Hz}$.

(d) Explain why photoelectric effect provides evidence for particle nature of light.

Worked solution

(a)

No photoelectrons ejected because photon energy is less than the work function. This could be because the UV radiation has frequency below the threshold frequency of the metal.

Answer: Frequency too low — photon energy $<$ work function

(b)(i)

Work function is the minimum energy required to release an electron from the surface of the metal.

Answer: Minimum energy needed to liberate an electron

(b)(ii)

Maximum KE of photoelectrons is given by $hf - \phi$. It depends only on frequency, not irradiance. Reducing irradiance reduces the number of photoelectrons, but their maximum KE stays the same.

Answer: No change to KE; fewer electrons emitted

(c)

For aluminium: threshold frequency $f_0 = 9.0 \times 10^{14}$ Hz. The line starts at $(f_0, 0)$ and increases linearly with slope h (same gradient as original graph). So graph shifts left compared with the original.

Answer: Straight line with same slope, intercept at 9.0×10^{14} Hz

(d)

Photoelectric effect shows that light transfers energy in discrete packets (photons). Emission of electrons depends on frequency, not intensity, which cannot be explained by wave theory. Only the particle (photon) model accounts for a threshold frequency and instantaneous emission.

Answer: Evidence that light consists of photons, not just waves

Final answers

(a) Photon energy < work function (frequency too low)

(b)(i) Minimum energy to liberate electron

(b)(ii) No change to KE; fewer electrons

(c) Graph line starts at 9.0×10^{14} Hz, slope h

(d) Shows light behaves as particles (photons)

Revision tips

- Work function ϕ is the minimum energy to free an electron.
- Einstein's photoelectric equation: $KE_{\text{max}} = hf - \phi$.
- Threshold frequency: $f_0 = \phi/h$.
- Intensity affects number of electrons, not KE.
- Photoelectric effect supports particle model of light.